QQ-A-250/13E
January 28, 1971
SUPERSEDING
Fed. Spec. QQ-A-250/13D
March 17, 1967

FEDERAL SPECIFICATION

ALUMINUM ALLOY ALCLAD 7075, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy alclad 7075 plate and sheet; the general requirements are covered in QQ-A-250. The plate and sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (7075) core with thin layers of an aluminum alloy (7072) anodic to the core and of approximately equal thickness bonded to both surfaces.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, T6, T62, T651, or F temper, as specified (see 6.2). The definitions of these tempers shall be as specified in American National Standards Institute ANSI Std. H35.1.

2. APPLICABLE DOCUMENTS

Latest issue of Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC. 20402.

FSC 9535

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T12:04Z
Check the source to verify that this is the current version before use.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specification:

MIL-I-8950 - Inspection, Ultrasonic, Wrought Metals, Process for.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

American National Standards Institute ANSI Standard:

H35.1 - Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the alclad plates and sheets shall conform to the requirements specified in table I for core and cladding, respectively.

TABLE 1. Chemical composition 1/

Element	Analysis			
	Core (7075)		(Cladding (7072)	
	Minimum	Maximum	Minimum	Maximum
	Percent	Percent	Percent	Percent
Zinc	5.1	6.1	0.8	1.3
Magnesium	2.1	2.9	-	0.10
Copper	1.2	2.0	-	.10
Chromium	0.18	0.35	-	-
Manganese	-	0.30	-	.10
Iron	-	0.50	-	2/
Silicon	-	0.40	-	2/
Titanium	-	.20	-	-
Others, each	-	.05	-	.05
Others, total	-	.15	-	.15
Aluminum	Remainder		Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the above table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

2/ Iron plus silicon 0.7 percent maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

TABLE II. Mechanical properties

Temper	Thickness	Tensile strength, minimum	Yield strength at 0.2 percent offset or at extension indicated		Elongation in 2 in. or 4 times D 1/, 2/, minimum
			Minimum	Extension under load	
	Inches	P.s.i.	P.s.i.	Inch/Inch	Percent
O	0.008 thru 0.014	36,000 3/	20,000 3/	0.0041	9
	.015 thru .062	36,000 3/	20,000 3/	.0041	10
	.063 thru .187	38,000 3/	20,000 3/	.0040	10
	.188 thru .499	39,000 3/	21,000 3/	.0041	10
	4/ .500 thru 1.000	40,000 3/	-	-	10
T6 and T62 5/	0.008 thru 0.011	68,000	58,000	0.0076	5
	.012 thru .039	70,000	60,000	.0078	7
	.040 thru .062	72,000	62,000	.0080	8
	.063 thru .187	73,000	63,000	.0081	8
	.188 thru .249	75,000	64,000	.0082	8
T651 and T62 5/	0.250 thru 0.499	75,000	65,000	0.0083	9
	4/ .500 thru 1.000	78,000	68,000	.0086	7
	4/ 1.001 thru 2.000	77,000	67,000	.0085	6
	4/ 2.001 thru 2.500	76,000	64,000	.0082	5
	4/ 2.501 thru 3.000	72,000	61,000	.0079	5
	4/ 3.001 thru 3.500	71,000	58,000	.0076	5
	4/ 3.501 thru 4.000	67,000	54,000	.0072	3
F	All	6/	6/	6/	6/

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.
3/ Maximum
4/ The properties for these thicknesses are those of the core alloy since the tests are made on a round specimen machined from the plate.
5/ Material in the T62 temper is not available from the materials producers.
6/ No requirements.

3.2.2 Mechanical properties after heat treatment. In addition to conforming to the requirements of 3.2.1, material in the annealed (O) and the as-fabricated (F) tempers shall, after proper solution heat-treatment and aging, also conform to the requirements of table II for the T6 and T62 tempers. Material as received in the T6 and T651 tempers shall, after proper re-solution heat-treatment and aging, be capable of conforming to the requirements specified in table II for the T6 and T62 tempers.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:04Z
Check the source to verify that this is the current version before use.

3.2.3 Bend test. Bend-test specimens taken from material shall be capable of withstanding, without cracking, the bend test specified in QQ-A-250. The values for bend factor N are given in table III.

TABLE III. Bend test factor "N"

Thickness of material	Tempers	
	O	T6 and T62
Inch		
0.008 thru 0.020	1	6
.021 thru .032	1	7
.033 thru .063	2	7
.064 thru .091	3	8
.092 thru .125	3	9
.126 thru .249	4	10
.250 thru .499	6	12

3.3 Cladding thickness.

3.3.1 Thickness of cladding plates. The aluminum alloy (7072) plates that are bonded to the two sides of the aluminum alloy (7075) ingot or slab to form a composite that is to be rolled to the finished thickness shall each have a thickness as specified in table IV.

TABLE IV. Cladding thickness

Thickness of finished plate or sheet	Nominal cladding thickness per side; percent of composite thickness	Average thickness per side of cladding on finished plate or sheet; minimum percent of plate or sheet thickness
Inches		
Under 0.063	4	3.2
0.063 thru 0.187	2.5	2
0.188 and over	1.5	1/ 1.2

1/ For plate 0.500 and over, the average cladding thickness per side shall have a maximum value of 3 percent of the plate thickness.

3.3.2 Thickness of cladding. If question arises concerning the thickness of cladding of the finished sheet or plate, samples examined in accordance with QQ-A-250 shall show an average of cladding on each side, not less than that specified in table IV.

3.4 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

TABLE V. Ultrasonic discontinuity acceptance limits 1/

Thickness	Maximum weight per piece	Discontinuity class 2/
Inches	Pounds	
0.500 thru 1.499	2,000	B
1.500 thru 3.000	2,000	A
3.001 thru 4.000	2,000	B

1/ Discontinuities in excess of those listed in table V may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in non-critical areas.
2/ The discontinuity class limits are defined in MIL-I-8950.

3.5 Marking. In addition to the marking required in Fed. Std. No. 184, material in the T6 and T651 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-250.)

4.1 Mechanical tests after heat treatment.

4.1.1 Number of tests, after heat treatment. From material in the annealed (O) and as-fabricated (F) tempers, an additional number of specimens equal to that required by QQ-A-250 shall be taken and tested after solution heat treatment and artificial aging to determine compliance with 3.2.2.

5. PREPARATION FOR DELIVERY
(See QQ-A-250.)

6. NOTES

6.1 Intended use. Alclad 7075 plates and sheets are used where very high strength with good resistance to corrosion are required. Formability is not as good as for lower strength aluminum alloys, cost is higher, and ductility may be somewhat lower.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Form and temper required (see 1.2.1).

(c) Dimensions required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

(d) Requirements for sizes not specifically covered (See QQ-A-250).

(e) Whether ultrasonic inspection of plate is required (see 3.4).

(f) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Temper. Temper T651 is available in plate only.

MILITARY CUSTODIANS:
Army - MR
Navy - AS
Air Force - 11

Review activities:
Army - MR, MI, MU, WC
Navy - SH, AS, YD, EC
Air Force - 85

User activities:
Army - EL, ME
Navy - MC, OS

Preparing activity:
Navy - AS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards
Department (ESSD) Code 93
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE 1973—714-866/3883 2-1

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:04Z
Check the source to verify that this is the current version before use.